CHAPTER 20

Ruby and the Web

There's a good chance that if you are reading this book, you are intending to use Ruby in the context of some kind of Web application. Ruby is used as the language for a lot of web tools, not just Ruby on Rails,[1] but also many other web frameworks and third-party tools.

While the Ruby ecosystem is full of web tools, most of those tools are third-party gems and not part of the core Ruby distribution. Core Ruby does provide an implementation of the Common Gateway Interface (CGI), which was the original dynamic web standard, but as we write this in 2022, you are unlikely to be writing CGI scripts directly, so we're not going to spend time on CGI in this book.

Many options are available for using Ruby to implement web applications, and a single chapter can't do them all justice. Instead, we'll touch on some of the highlights and point you toward libraries and resources that can help. In particular, we'll focus on:

- The web-related utilities that are part of the Ruby Standard Library
- ERB, the most common third-party tool for templating
- Rack, a third-party standard for all behavior common to Ruby web frameworks
- Sinatra, perhaps the simplest of the powerful Ruby web frameworks
- Puma, the most popular Ruby web server

Ruby's Web Utilities

The Ruby standard distribution includes some core utilities as part of the `CGI` class and `CGI::Util` module.

CGI Encoding

When dealing with URLs and HTML code, you must be careful to quote certain characters. For instance, a slash character (/) has special meaning in a URL, so it must be "escaped" if it's not part of the path name. That is, any / in the query portion of the URL will be translated to the string `%2F` and must be translated back to a / for you to use it. Space and ampersand are also special characters.

To handle this, `CGI` provides the methods `escape` and `unescape`:

```
require "cgi"
```

1. http://www.rubyonrails.org